

Boyue Du

UCLA ECE | NVIDIA Hackathon Winner | AI & Machine Learning
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Education

University of California, Los Angeles (UCLA) <i>Master of Science in Electrical and Computer Engineering</i>	Los Angeles, CA <i>Sep. 2025 – Jun. 2027 (Expected)</i>
University of Wisconsin–Madison <i>Bachelor of Science in Electrical Engineering GPA: 3.7/4.0</i>	Madison, WI <i>Sep. 2021 – May 2025</i>

Technical Skills

Programming:	Python, Java
AI & ML:	LLMs, Generative AI, Multimodal AI, AI Agents, Model Compression, Hardware-Aware ML
Tools:	PyTorch, LangChain, REST APIs, Flask, Supabase, Linux, CUDA, Git

Selected AI Projects

Symphonia – Culture Impact Track Winner, NVIDIA GTC Hackathon [GitHub] <i>NVIDIA, sponsored by HP</i>	2026 <i>San Jose, CA</i>
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- Built a **multimodal AI application** for classical-music discovery that combines audio analysis, **LLM-based contextual explanations**, personalized recommendations, guided listening, and interactive visualizations.
- Engineered the **end-to-end Flask application and LLM workflow**, integrating **GPU-accelerated generative modules** and iteratively improving output relevance, consistency, and user experience through prompt and demo evaluation.

Vision-Language-Action (VLA) Model Evaluation for Autonomous Driving <i>Mobility Lab, UCLA</i>	2026 <i>Los Angeles, CA</i>
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- Benchmarked **NVIDIA Alpamayo-R1** on **1000+** driving scenarios, conducting controlled experiments on temperature, top-p, and history length to quantify trajectory prediction performance using minADE.
- Identified a **14.6%** increase in minADE with longer temporal context, highlighting the model’s sensitivity to input history length.

Aletheia Research – Evidence-Grounded Research Agent [GitHub] <i>Independent Project</i>	2026 <i>Los Angeles, CA</i>
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- Built an **evidence-grounded LLM research agent** with an end-to-end pipeline for literature retrieval, source verification, iterative search, and citation-backed synthesis.
- Designed an **agent evaluation framework** across 9 real-world VLA research tasks, measuring coverage and evidence completeness and analyzing failure modes to identify retrieval and reasoning bottlenecks.

Research Experience

Muscle Synergy Representation Learning – NMF and Time-preserving Autoencoder <i>Project Lead & First Author, in collaboration with Beijing Normal University</i>	2026 <i>Remote</i>
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- Developed NMF and a time-preserving autoencoder for **low-dimensional representation learning** on 16-channel sEMG, modeling spatial muscle coordination and temporal activation across fatigue conditions.
- Designed a **cross-representation evaluation pipeline** using component alignment, cosine similarity, and permutation testing, demonstrating consistent fatigue-related patterns across linear and nonlinear representations.

Industry Experience

Business Developer [Media Coverage] <i>NaaS Technology</i>	Nov. 2023 – Jan. 2024 <i>Las Vegas, NV</i>
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- Supported North American business development for EV charging products and represented NaaS at **CES 2024**, translating technical capabilities into **customer-facing use cases**.

Awards

NVIDIA GTC Hackathon —Culture Impact Track Winner [HP Feature] [UCLA News]	2026
Silicon Valley Fellowship	2026
Novotny, Donald W. Scholarship, UW–Madison	2024
WV Distinguished Scholarship, UW–Madison	2023, 2024

Leadership & Credentials

Mentor	IEEE-Eta Kappa Nu (IEEE-HKN) , UCLA – Mentored undergraduate engineering students.
Graduate Reader	ECE 113 Digital Signal Processing, UCLA – Supported and graded a 200+ student DSP course.
Certifications	NVIDIA – Building LLM Applications with Prompt Engineering; AMD – ROCm Certification Associate.